From Derivations to Mathematical Prose

Here is a complete proof, as a textbook of analysis would print it, that a function uniformly continuous on a set A is continuous at every point of A:

> Let $x \in A$ and $\varepsilon > 0$. By uniform continuity, choose $\delta > 0$ (depending on ε only) such that $|f(u) - f(v)| < \varepsilon$ for all $u, v \in A$ with $|u - v| < \delta$. In particular $|f(x) - f(y)| < \varepsilon$ for every $y \in A$ with $|x - y| < \delta$, so this δ works for x and ε.

Section 6.4 proved the same fact as an eleven-line derivation (Example (Uniformly continuous implies continuous)), with three nested boxes: one for the point x_0, one for ε_0, one for the chosen δ_0. The paragraph has no boxes and no line numbers. *Where did the boxes go?* Nowhere. "Let $x \in A$ and $\varepsilon > 0$" opens the first two, "choose" opens the third, and the parenthesis "(depending on ε only)" records that δ came from a line that mentions ε and not x, which is exactly where uniform continuity and continuity differ. Words such as "let", "choose", "suppose" and "in either case" are the boxes, spoken. This chapter makes that correspondence exact in both directions. Learning to hear the boxes is what lets a reader working alone check a proof, by restoring the derivation underneath wherever a step is in doubt; learning to speak them is what makes a proof checkable by someone else.

The chapter builds a dictionary from rules to phrases, writes two proofs at four levels from full derivation to plain prose, fixes once and for all which steps plain prose may leave unsaid, and turns all of this into habits for reading, writing and checking proofs, with a catalogue of the errors that recur. From *Analysis on the Real Line*, Chapter 1 on, proofs are written in plain prose, and a rule tag appears only where the rule is itself the point, as in a review exercise that looks back at this volume; *Sets, Relations, and Cardinality*, Chapters 1–3 may still tag some proofs. The chapter ends with the one rule that mathematicians have seriously argued about: reductio ad absurdum, the rule behind excluded middle, what analysis would lose without it, and where this book stands.

What you need. Chapters 1–7. From Chapters 1 and 2: the rules for every connective, derivations with their boxes and scope rule (Definition 1.4.2), derived rules (Definition 2.3.1), the classical rules of Section 2.5 and their interderivability (Theorem 2.5.7), the cut rule (Lemma 2.3.3), and the two running examples Theorem 2.1.3 and Theorem 2.2.1. From Chapter 4: induction (Axiom 4.2.1), truth tables and refuting rows (Example (Working backwards from F), Definition 4.5.3, Corollary 4.6.3), soundness (Theorem 4.6.2), completeness and the proof of Lemma 4.7.3 (Theorem 4.7.4, Corollary 4.7.5), and, for Section 8.7 only, the optional Theorem 4.6.4. From Chapters 5 and 6: the quantifier rules and their side conditions (Definition 5.4.1, Definition 5.6.1), the bounded rules (Proposition 6.1.1), definitions of analysis translated into formulas (Section 6.1), the laws for negating quantifiers (Proposition 6.2.1), and the derivation that uniform continuity implies continuity together with the place where its converse breaks down (Example (Uniformly continuous implies continuous), Example (The search from C to UC), and the counter-model Example (The square function on the real line)). School arithmetic and the order of $\mathbb{R}$ are used in examples as facts.

Time. About 35 hours on the core track (all of A, one B and one C per section, the exit checks); about 45 hours for everything.

8.-1.1 Roadmap

- **The connective rules in prose.** The dictionary rows for the connectives and the classical rules, each with an analysis proof annotated phrase by phrase; the three faces of "proof by contradiction"; and the contradiction proofs that are direct or contrapositive proofs in costume.
- **The quantifier rules in prose.** The rows for "for every", "there exists", equality and unique existence, and for induction, "it suffices to show" and "without loss of generality"; the dependency parenthesis; and the quantifiers that English leaves unwritten or puts in the wrong place.
- **Four levels of one proof.** Full derivation, skeleton, annotated prose and plain prose; the convention that says which steps plain prose may leave silent; the irrationality of $\sqrt{2}$ and "uniformly continuous implies continuous", each written four times; and how to check a doubtful sentence by expanding it.
- **Reading a proof: self-explanation.** The book's training passage for the habit of naming, at each line, what makes it true; four questions to ask of every line; and five strategies that good readers apply to a proof as a whole.
- **Writing a proof.** A seven-step procedure from the statement to plain prose, the conventions of written mathematics as rules made audible, and three worked examples: labeling a proof, turning a derivation into a paragraph, and writing a new proof from its goal.
- **Checking proofs, and what is not allowed.** Two questions for every line, the catalogue (N1)–(N11) of errors with the rule each one breaks, and four student proofs checked line by line.
- **The dispute over excluded middle.** Intuitionistic logic as the rules without reductio; why excluded middle cannot be refuted; what a proof provides under the BHK reading; an irrational power of an irrational number that is rational, with and without a witness; a short history; what analysis would lose; proofs as programs; and the book's position.

8.-1.2 Named moves

- **Hear the rule in the phrase.** "Let $x \in A$" opens a $\forall$I box and "suppose" an assumption; "choose" is $\exists$E; "take" and "let $\delta := \ldots$" are $\exists$I; "suppose, for contradiction" is RAA or $\neg$I; both boxes end in $\bot$, and what the box assumes and the line written after it closes tell which.
- **Expand to check.** When a sentence is doubtful, rewrite that stretch, and only that stretch, as a skeleton, and look for the line that breaks a rule.
- **Skeleton first.** Write the first and the last sentence of every box before filling any.
- **Self-explain each line** by earlier lines, definitions and named results, and the rule that combines them; never by paraphrase.
- **Two columns when checking:** *is the line true?* and *does it follow, and by which rule?* The first column finds where to look; the second decides.
- **Say what depends on what.** Every chosen object carries its dependencies: the names in the line it comes from, among the boxes open when it was chosen.